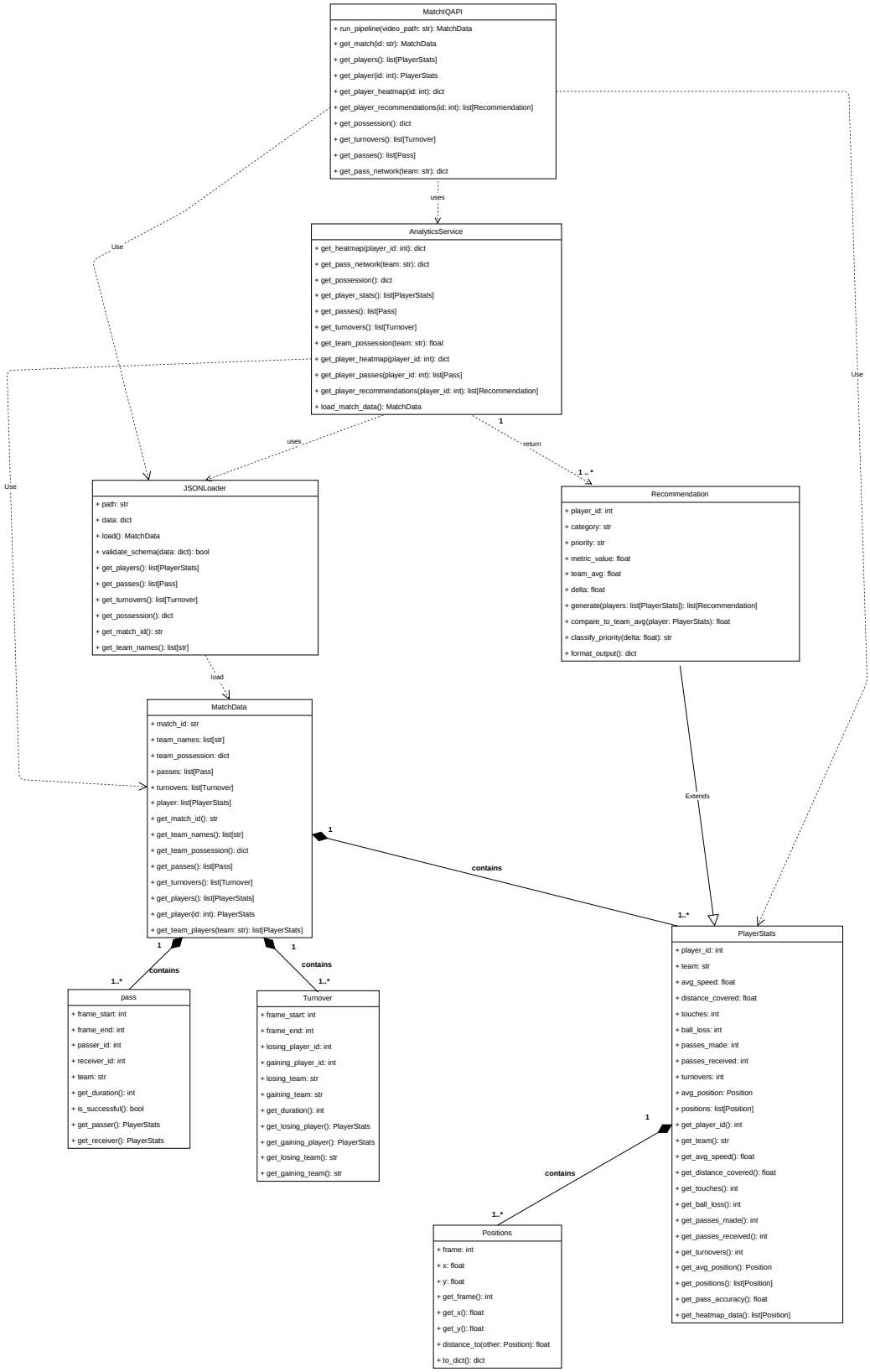
MatchIQAPI
+ run_pipeline(video_path: str): MatchData
+ get_match(id: str): MatchData
+ get_players(): list[PlayerStats]
+ get_player(id: int): PlayerStats
+ get_player_heatmap(id: int): dict
+ get_player_recommendations(id: int): list[Recommendation]
+ get_possession(): dict
+ get_turnovers(): list[Turnover]
+ get_passes(): list[Pass]
+ get_pass_network(team: str): dict
uses
Use
AnalyticsService
+ get_heatmap(player_id: int): dict
+ get_pass_network(team: str): dict
+ get_possession(): dict
+ get_player_stats(): list[PlayerStats]
+ get_passes(): list[Pass]
+ get_turnovers(): list[Turnover]
+ get_team_possession(team: str): float
+ get_player_heatmap(player_id: int): dict
+ get_player_passes(player_id: int): list[Pass]
+ get_player_recommendations(player_id: int): list[Recommendation]
+ load_match_data(): MatchData
Use
Use
uses
1
return
1 ..*
JSONLoader
+ path: str
+ data: dict
+ load(): MatchData
+ validate_schema(data: dict): bool
+ get_players(): list[PlayerStats]
+ get_passes(): list[Pass]
+ get_turnovers(): list[Turnover]
+ get_possession(): dict
+ get_match_id(): str
+ get_team_names(): list[str]
Recommendation
+ player_id: int
+ category: str
+ priority: str
+ metric_value: float
+ team_avg: float
+ delta: float
+ generate(players: list[PlayerStats]): list[Recommendation]
+ compare_to_team_avg(player: PlayerStats): float
+ classify_priority(delta: float): str
+ format_output(): dict
load
Extends
MatchData
+ match_id: str
+ team_names: list[str]
+ team_possession: dict
+ passes: list[Pass]
+ turnovers: list[Turnover]
+ player: list[PlayerStats]
+ get_match_id(): str
+ get_team_names(): list[str]
+ get_team_possession(): dict
+ get_passes(): list[Pass]
+ get_turnovers(): list[Turnover]
+ get_players(): list[PlayerStats]
+ get_player(id: int): PlayerStats
+ get_team_players(team: str): list[PlayerStats]
1
contains
1..*
1
contains
1..*
1
contains
1..*
pass
+ frame_start: int
+ frame_end: int
+ passer_id: int
+ receiver_id: int
+ team: str
+ get_duration(): int
+ is_successful(): bool
+ get_passer(): PlayerStats
+ get_receiver(): PlayerStats
Turnover
+ frame_start: int
+ frame_end: int
+ losing_player_id: int
+ gaining_player_id: int
+ losing_team: str
+ gaining_team: str
+ get_duration(): int
+ get_losing_player(): PlayerStats
+ get_gaining_player(): PlayerStats
+ get_losing_team(): str
+ get_gaining_team(): str
PlayerStats
+ player_id: int
+ team: str
+ avg_speed: float
+ distance_covered: float
+ touches: int
+ ball_loss: int
+ passes_made: int
+ passes_received: int
+ turnovers: int
+ avg_position: Position
+ positions: list[Position]
+ get_player_id(): int
+ get_team(): str
+ get_avg_speed(): float
+ get_distance_covered(): float
+ get_touches(): int
+ get_ball_loss(): int
+ get_passes_made(): int
+ get_passes_received(): int
+ get_turnovers(): int
+ get_avg_position(): Position
+ get_positions(): list[Position]
+ get_pass_accuracy(): float
+ get_heatmap_data(): list[Position]
1
contains
1..*
Positions
+ frame: int
+ x: float
+ y: float
+ get_frame(): int
+ get_x(): float
+ get_y(): float
+ distance_to(other: Position): float
+ to_dict(): dict